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Recommender Systems

User Behavior and Data

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Definitions and Implications

- Information retrieval (IR) … is the process of obtaining information system resources that are relevant to an information need from a collection of those resources. … Information retrieval is the science of searching for information in a document, searching for documents themselves, and also searching for the metadata that describes data, and for databases of texts, images or sounds. [Wikipedia]
- Implication: user has a specific information need, typically represented as a query; system tries to find the most relevant items w.r.t. query

https://en.wikipedia.org/wiki/Information_retrieval
Slide adapted from Learning from User-Generated Data, Markus Schedl, Peter Knees, JKU Linz, 2021.

Definitions and Implications

- Browsing: interactive task in which the user is more interested in exploring the document collection than in retrieving documents which satisfy a specific information need [Baeza-Yates and Ribeiro-Neto, 1999]
- Implication: user has an undirected information need (hard to express as a query); system should provide a good interface for discovering items

Ricardo Baeza-Yates and Berthier Ribeiro-Neto. 2011. Modern Information Retrieval: The concepts and technology behind search (2nd. ed.). Addison-Wesley Publishing Company, USA.
Slide adapted from Learning from User-Generated Data, Markus Schedl, Peter Knees, JKU Linz, 2021.

Definitions and Implications

- A recommender system, or a recommendation system [...], is a subclass of information filtering system that seeks to predict the “rating” or “preference” a user would give to an item. [Wikipedia]
- Implication: the system searches for potentially relevant items based on user’s actions or preferences that form a user profile (can, e.g., be observed during retrieval or browsing)

https://en.wikipedia.org/wiki/Recommender_system
Slide adapted from Learning from User-Generated Data, Markus Schedl, Peter Knees, JKU Linz, 2021.

Required Involvement of the User

- Retrieval: user has a specific information need, typically represented as a query; system tries to find the most relevant items wrt. Query
- Browsing: user has an undirected information need; system should provide a good interface for discovering items
- Recommending: the system searches for potentially relevant items based on user's actions or preferences (e.g., observed during retrieval or browsing)

Required Involvement of the User

- Retrieval: user has a specific information need, typically represented as a query; system tries to find the most relevant items wrt. Query ➜ User needs to be active
- Browsing: user has an undirected information need; system should provide a good interface for discovering items ➜ User can be (almost) passive
- Recommending: the system searches for potentially relevant items based on user's actions or preferences (e.g., observed during retrieval or browsing) ➜ User needs to be active

Collecting Evidence for User Taste

Data Processing Model

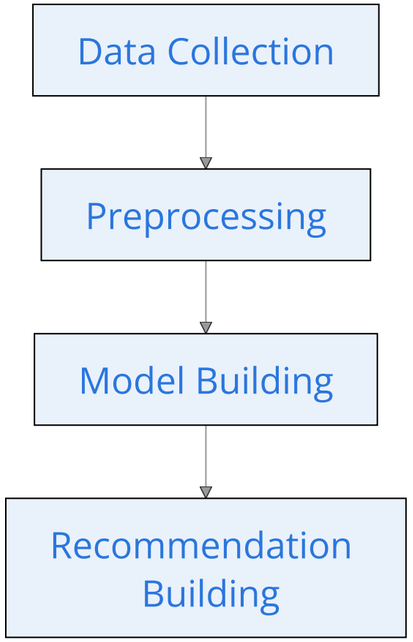


Figure adapted from Practical Recommender Systems: Kim Falk; Manning Publications, 2019.

Evidence for User Taste

- Collect events and behavior that provide an indication of the user's tastes.
- Evidence collected about a user informs the user profile used by the RecSys.
- Depending on the domain and platform, evidence can either be collected implicitly or explicitly.

Observational Nature of Recommender Data

We observe:

- clicks
- views
- purchases
- ratings

We do **not** observe what users would have chosen if exposed differently

Logged Interaction Data ≠ Ground Truth Preferences

Recommender systems learn from **observational data**, not controlled experiments.

Observational Nature of Recommender Data

Core problem: Exposure is **not random**.

$$P(\text{interaction}) = P(\text{interaction} \mid \text{exposed}) \cdot P(\text{exposed})$$

Consequences of Observational Data

Offline evaluation assumes data is independent and identically distributed (i.i.d.) and that the observed interactions reflect true preferences, but:

Missing-Not-at-Random (MNAR)

Unobserved ≠ negative

Exposure Bias

Users can only interact with shown items

Feedback Loops

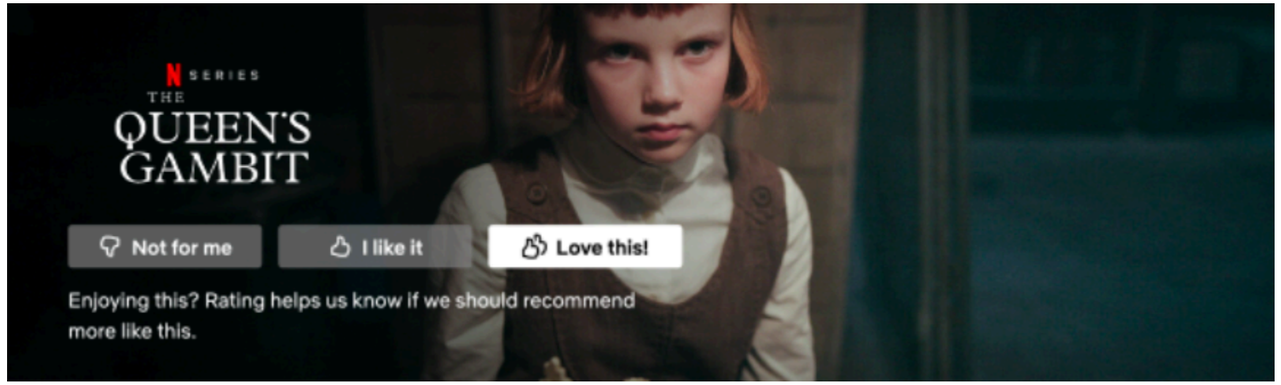
Model ➜ Exposure ➜ Interaction ➜ Retraining

This motivates counterfactual evaluation later.

Explicit User Data: Ratings

- Binary ratings (thumbs up/down)
- Unary ratings (“Like”)
- “Continuous” slider and/or text box
- Likert scale: discrete steps (e.g., 5 or 10 stars, possibly in half steps)

Explicit User Data: Ratings

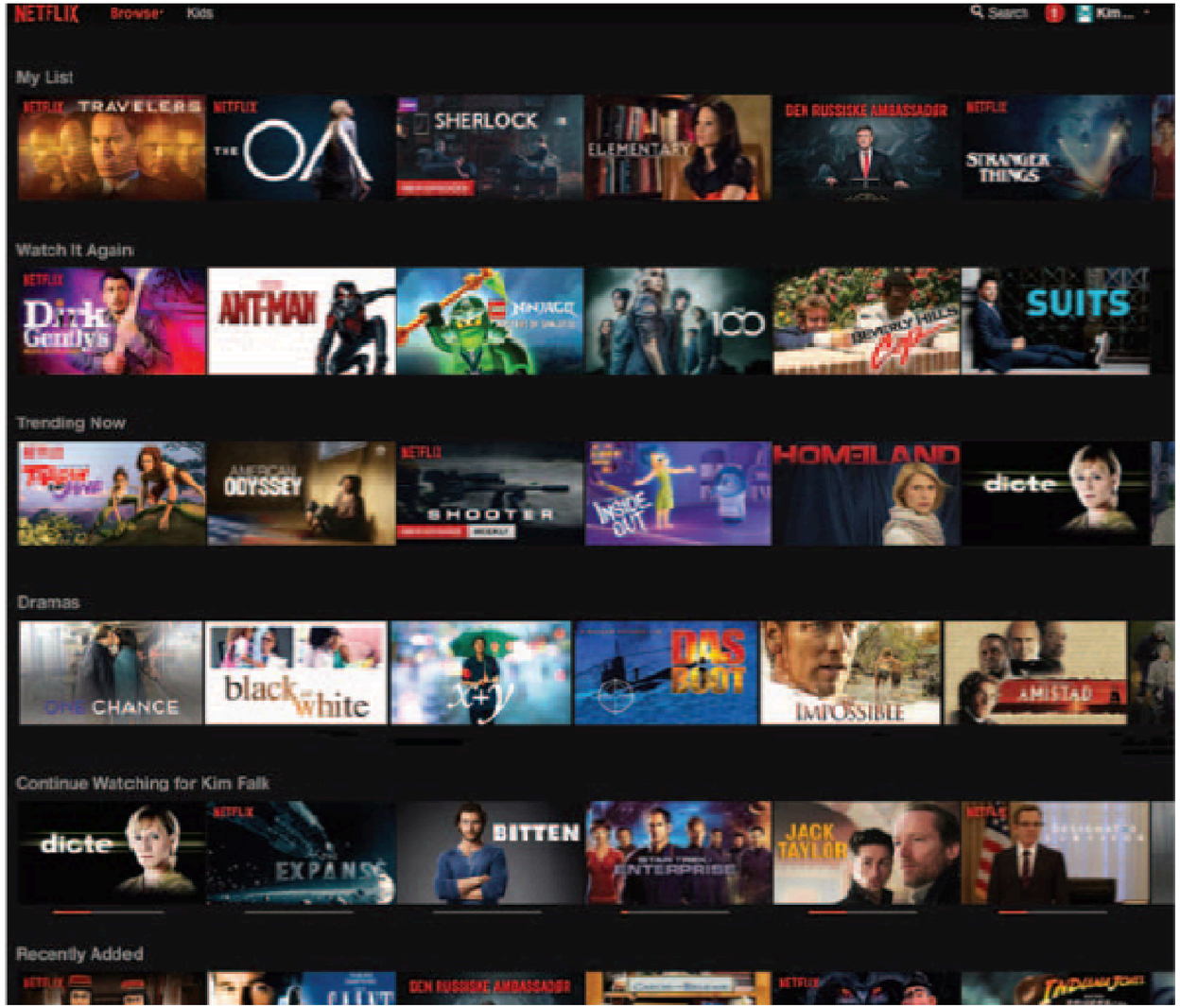


Implicit User Data

- Record user activity on the platform.
- Example
  - Click-stream / browsing behavior
  - View time
  - Consumption time
  - Items added to favorites, shopping lists, etc.
  - Items skipped in stream of items (e.g., music streaming)
  - …

Implicit User Data: Netflix Showcase

The first personalized row



Showcase taken from Practical Recommender Systems: Kim Falk; Manning Publications, 2019.

Implicit User Data: Netflix Showcase

- (Hypothetical) user Jimmy (ID 1234) on a Saturday night:
  - Scrolls the Drama (ID 2) row
  - Hovers the mouse over a movie (ID 41335) to get details
  - Clicks to get more details about the movie (ID 41335)
  - Starts watching the movie (ID 41335)

Implicit User Data: Netflix Showcase

- Evidence gathered by Netflix about Jimmy

Event	Meaning	EventName
Scrolling a themed row	User is interested in the theme, here: Drama.	genreView
Placing the mouse over a film to request an overview of content	User is interested in the movie (a drama), thereby showing interest in this category.	Details
Clicking the film to request the details of content	User is more interested in the movie.	moreDetails
Adding the film to My list	User intends to watch the movie later.	addToList
Starting to watch the film	User “purchases” the movie.	playStart

Implicit User Data: Netflix Showcase

- Detailed log stored by Netflix

userID	contentID	Event	Date
1234	2	genreView	2021-02-07 20:01:00
1234	41335	Details	2021-02-07 20:02:21
1234	41335	moreDetails	2021-02-07 20:02:30
1234	41335	addToList	2021-02-07 20:02:55
1234	41335	playStart	2021-02-07 20:03:01

Implicit User Data

- Consumer/product relationship lifecycle:
  - Consumer browses
  - Consumer becomes interested in one or more products
  - Consumer adds product to basket or a list with intent to buy
  - Consumer buys products
  - Consumer consumes product
  - Consumer rates the product
  - Consumer resells/disposes/consumes product

Implicit User Data: Measures

- Page view
- Page duration (interpretation is highly domain-specific)

Duration	What it might mean (example)
< 5 sec	No interest
> 5 sec	Interested
> 1 min	Very interested
> 5 min	Probably went to get coffee
> 10 min	Interrupted or went away from the page without following a link

Implicit User Data: Measures

- Expansion Clicks
- Capture what the user is doing, better overview for customer

Implicit User Data: Measures

- Social media links (sharing)
- Save for later (favorites, wish list)
- Search terms
- Buy a product
- Consume a product – streaming
  - Start playing
  - Stop playing
  - Resume playing
  - Speeding
  - Playing it to the end
  - Replaying

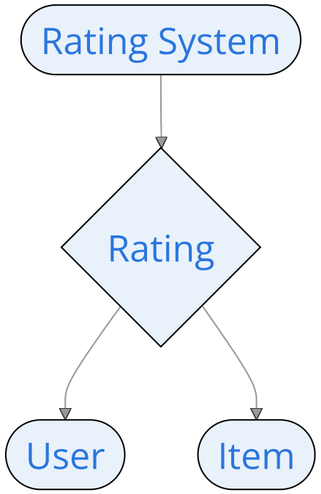
Explicit User Data: Ratings

- Explicit ratings by users
- Rating is highly user-specific
  - Critical users may never award five stars
  - Generous users may award five stars at high frequency
- Users do not always rate truthfully
- No rating != rating of 0 stars

User Ratings: Modeling

User Ratings

Rating: some form of feedback from the user (explicit or implicit)



User-Item Matrix

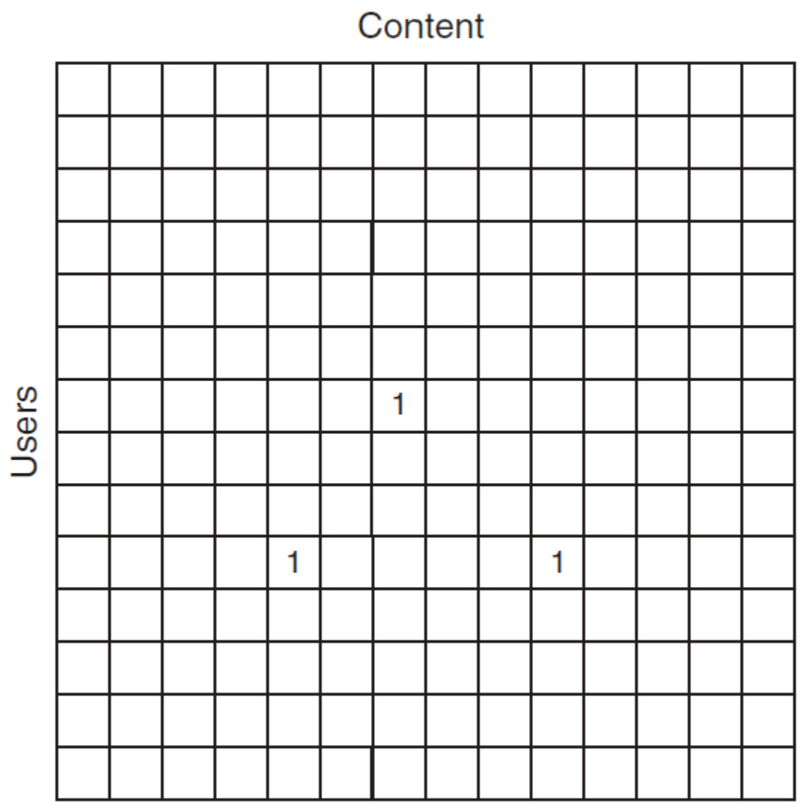
	Indiana Jones	Microcosmos	Avengers	Pete's Dragon
Sara	4	5		
Jesper	4		5	
Therese	5			3
Helle	4			5
Pietro		3	4	3
Ekaterina	3		3	0

Important

Difference between rating of 0 and empty cell!

Example taken from Practical Recommender Systems: Kim Falk; Manning Publications, 2019.

User-Item Matrix: Sparsity



- Sparsity usually > 99% for RecSys datasets
- Long tail

Figure taken from Practical Recommender Systems: Kim Falk; Manning Publications, 2019.

Cold Start Problem: Sparse or Missing Data

- **User Cold Start**: New user does not have any interaction history
- **Item Cold Start**: New item does not have any exposure or feedback
- **System Cold Start**: New platform does not have any data at all

Collaborative filtering struggles here, as it relies on user-item interactions to make recommendations.

This motivates: - Content-based methods - Hybrid models - Metadata & representation learning

User-Item Matrix: Explicit vs. Implicit

- Note that for filling the user-item matrix, we can user explicit and implicit feedback from users.
- In a later stage, we can further augment the scalar user-item interactions with further data on the user and the item (also: further implicit user data) .
- Multiple implicit data are often combined into a single rating via weighting.

User-Item Matrix: Implicit Ratings

- Buy events (binary)
- Often: scalar values (e.g., number of listening events) are binarized too (or: weighted using log-transformations)

	Indiana Jones	Microcosmos	Avengers	Pete's Dragon
Sara	1	1	0	0
Jesper	1	0	1	0
Therese	1	0	0	1
Helle	1	0	0	1
Pietro	0	1	1	1
Ekaterina	1	0	1	0

Note

No more empty cells

Ratings and their Relation to the RecSys Task

- Non-binary ratings: $f : U \times I \rightarrow \mathcal{R}$ can be modeled by a regression task.
  ➜ For each item predict the rating value (e.g., 0-5 stars).
- Binary ratings: $f : U \times I \rightarrow \mathcal{R}$ can be modeled by a classification task.
  ➜ For each item predict whether it will be in the 0 or in the 1 class.
- Choice of tasks also informs/determines the way the RecSys is evaluated (classification vs. regression metrics).

Ratings, Learning Tasks, and RecSys Objectives

We observe feedback:

$$f : U \times I \to \mathcal{R}$$

Depending on the feedback, we can model learning as:

- Regression (explicit ratings) predicts $\hat{r}_{ui}$ (Loss: MSE / MAE)
- Classification (binary/implicit) predicts $P(y_{ui} = 1)$ (Loss: cross-entropy)

Recommendation is a top-K ranking problem, not a regression or classification problem.

At inference time, we compute $\hat{i}_1, \ldots, \hat{i}_K = \arg\max_{i \notin I_u} s(u, i)$, where $s(u, i)$ is a scoring function that estimates the relevance of item i for user u.

Ratings, Learning Tasks, and RecSys Objectives cont.

- Regression / classification = *training formulation*
- Ranking = *decision objective*
- Problem: Low training loss does not necessarily guarantee good ranking performance (and thus good recommendations).
- Evaluation target (ranking metrics) is not the same as training target (regression/classification loss).

Ratings, Learning Tasks, and RecSys Objectives cont.

For user u, we aim to find a scoring function $s(u, i)$ such that the following holds for all pairs of items i^+ and i^-:

$$s(u, i^+) > s(u, i^-)$$

where:

- i^+ = preferred / interacted item
- i^- = non-preferred item

Recommendation is fundamentally about **relative ordering**, not absolute score accuracy.

Ratings, Learning Tasks, and RecSys Objectives cont.

Different loss formulations approximate the ranking objective differently.

- **Pointwise**: Treat prediction independently.
  - Regression (MSE), Classification (cross-entropy)
  - Simple, indirectly optimizes ranking
- **Pairwise**: Optimize relative ordering of pairs of items.
  - BPR (Bayesian Personalized Ranking)
  - More direct optimization of ranking
- **Listwise**: Optimize ranking metrics directly.
  - nDCG-based losses
  - Closest to evaluation objective, complex optimization